

Motivation: high dimensional data analysis via PCA

- Suppose you measure N variables under P experimental conditions
 ie firing rates of N neurons over P trials
 or expression pattern of N genes over P conditions

• this yields an $N \times P$ data matrix $X = \begin{bmatrix} | & | & | & \dots & | \\ & & & & \\ & & & & \end{bmatrix}$

- which then yields an empirical covariance matrix

$$\hat{C} = \frac{1}{P} X X^T$$

- PCA examines eigenvalues and eigenvectors of $\hat{C}$



- often data covariance matrices have a noise bulk and outlier signal eigenvalues
- but what is the boundary between signal and noise?
- need a null model of data

Null model for PCA

- assume data is iid noise of variance 1

$$X_{iM} \sim \mathcal{N}(0, 1) \quad \text{for } i=1 \dots N \quad M=1 \dots P$$

- then the empirical covariance matrix $\hat{C} = \frac{1}{P} X X^T$ is a random matrix
- what does its spectrum, ie the spectrum of noise look like?

- two limits:

classical low
dim statistics
limit

$$N \sim \mathcal{O}(1)$$

$$P \rightarrow \infty$$

$$\hat{C} \rightarrow I$$

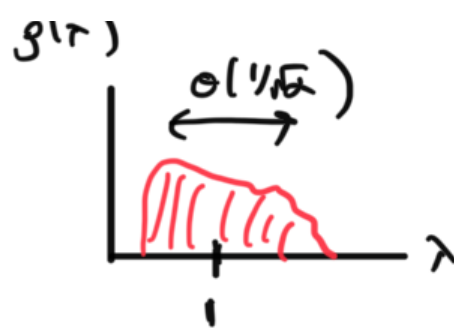
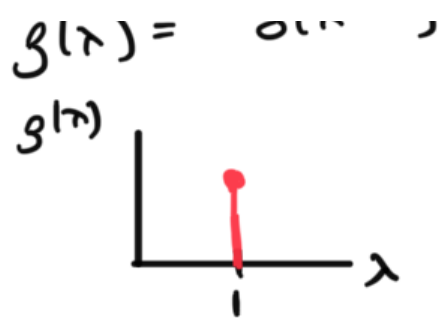
modern high dim
statistics limit

$$N \rightarrow \infty$$

$$P \rightarrow \infty$$

$$\alpha = P/N \sim \mathcal{O}(1)$$

measurement density =
measurements per dimension



- for high dimensional noise, the noise ball will appear elliptical with large and small directions
- as moment density increases, noise ball becomes isotropic and $g(\lambda)$ collapses

Goal: compute shape of $g(\lambda)$ as a function of α :

use covg method $\rightarrow$ stieljes transform $\rightarrow g(\lambda)$

$$\mathcal{E}_X(z) = \frac{1}{N} \text{Tr} [zI - \hat{C}]^{-1}$$

$$\mathcal{E}_X(z) = \frac{1}{N} \text{Tr} \left[zI - \frac{1}{P} X X^T \right]^{-1}$$

Step 1: compute a self consistent equation for $\mathcal{E}(z) = \langle \mathcal{E}_X(z) \rangle_X$ via covg method

Step 2: solve this equation to get $\mathcal{E}(z)$

Step 3: use inversion formula to get $g(\lambda)$

$$g(\lambda) = \frac{1}{\pi} \lim_{\epsilon \rightarrow 0} \text{Im} \mathcal{E}(\lambda - i\epsilon)$$

Step 1 covg method to get self consistent eqn for $\mathcal{E}(z)$

Let $M = zI - \frac{1}{T} X X^T$ $E \equiv M^{-1}$ $\mathcal{E}_X(z) = \frac{1}{N} \text{Tr} E$

$$M \equiv \left[\begin{array}{c|c} M_{11} & M_{12} \\ \hline M_{21} & M_{22} \end{array} \right] \quad E \equiv \left[\begin{array}{c|c} E_{11} & E_{12} \\ \hline E_{12} & E_{22} \end{array} \right]$$

$$E_{11}^{-1} = M_{11} - M_{12} M_{22}^{-1} M_{21}$$

Schur's complement formula

apply to

$$\begin{array}{c} 1 \\ 2 \\ 3 \\ \vdots \end{array} \left[\begin{array}{c|cccc} & 1 & 2 & 3 & \dots & N \\ \hline & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \end{array} \right]$$

$$M_{11} = z - \frac{1}{T} \sum_{m=1}^T x_{1m} x_{m1}$$

$$(M_{12})_i = - \frac{1}{T} \sum_{m=1}^T x_{1m} x_{mi} \quad i=2, \dots, N$$

$$\frac{1}{E_1(z)} = \underbrace{M_{11}} - \underbrace{M_{12} M_{22}^{-1} M_{21}}$$

$$z = \underbrace{\frac{1}{P} \sum_{m=1}^P X_{1m} X_{m1}}_{\rightarrow 1 \text{ as } T \rightarrow \infty}$$

$$\sum_{m,r=1}^P \sum_{i,j=2}^N \underbrace{\left[\frac{1}{P} X_{1m} X_{mi} \right]}_{\text{from } (M_{12})_i} (M_{22}^{-1})_{is} \underbrace{\left[\frac{1}{P} X_{jr} X_{r1} \right]}_{\text{from } (M_{12})_s}$$

$$= \frac{1}{P} \sum_{m,r=1}^P X_{1m} \underbrace{\left[\frac{1}{P} \sum_{i,j=2}^N X_{mi} (M_{22}^{-1})_{is} X_{jr} \right]}_{\equiv A_{mr}}$$

$$\equiv A_{mr}$$

a $P \times P$ matrix

note: we grouped terms together this way because:

X_{1m} for $m=1 \dots P$ is independent of
 X_{im} for $i=2 \dots N$
 and $(M_{22})_{is}$ for $i,s=2 \dots N$ } ie indep of A_{mr}

$$\text{as } P \rightarrow \infty \simeq \frac{1}{P} \sum_{m,r=1}^P \langle X_{1m} A_{mr} X_{r1} \rangle X_{1m}$$

$$= \frac{1}{P} \sum_{m,r=1}^P \langle X_{1m} X_{r1} \rangle_{X_{1m}} A_{mr}$$

$$= \frac{1}{P} \sum_{m,r=1}^P \delta_{mr} A_{mr}$$

$$= \frac{1}{P} \sum_{m=1}^P A_{mm}$$

$$= \frac{1}{P} \text{Tr } A$$

$$= \frac{1}{P} \sum_{i,j=2}^N \frac{1}{P} \sum_{m=1}^P X_{mi} (M_{22}^{-1})_{is} X_{jm}$$

$$= \frac{1}{P} \sum_{i,j=2}^N \left[\frac{1}{P} \sum_{m=1}^P X_{mi} X_{mj} \right] (M_{22}^{-1})_{is}$$

$$= \frac{1}{P} \text{Tr } \hat{C}^{N-1} M_{22}^{-1}$$

$$= \frac{1}{P} E_1(z)$$

$$= \frac{1}{P} \text{Tr} \hat{C}^{N-1} E_{N-1}(z)$$

resolvent matrix of cavity system

empirical covariance of cavity system

we need to extract Stieltjes transform of cavity system

$$E_{N-1}(z) = \frac{1}{N-1} \text{Tr} E_{N-1}(z)$$

$$\text{ndc } \text{Tr} \hat{C}^{N-1} E_{N-1}(z)$$

$$= \text{Tr} \hat{C}^{N-1} [zI - \hat{C}^{N-1}]^{-1}$$

$$= \text{Tr} [\hat{C}^{N-1} - zI + zI] [zI - \hat{C}^{N-1}]^{-1}$$

$$= \text{Tr} [\hat{C}^{N-1} - zI] [zI - \hat{C}^{N-1}]^{-1} + z \text{Tr} [zI - \hat{C}^{N-1}]^{-1}$$

$$= -\text{Tr} I_{N-1} + z \text{Tr} [zI - \hat{C}^{N-1}]^{-1}$$

$$= -(N-1) + z (N-1) E_{N-1}(z)$$

Stieltjes transform of cavity system

Now put it all together:

$$\frac{1}{E_{11}(z)}$$

=

$$\frac{M_{11}}{z-1}$$

$$\frac{1}{E_{11}(z)}$$

=

$$\frac{1}{E_{11}(z)}$$

take expectation value

$$- \frac{M_{12} (M_{22}^{-1}) M_{12}}{P}$$

$$- \frac{N-1}{P} [-1 + z E_{N-1}(z)]$$

$$\frac{1}{\alpha} = \frac{N}{P} \text{ at large } N$$

$$E(z) \text{ at large } N$$

$$\left\langle \frac{1}{E_{11}(z)} \right\rangle \approx \frac{1}{\langle E_{11}(z) \rangle}$$

$$= \frac{1}{\frac{1}{N} \sum_{i=1}^N \langle E_{ii}(z) \rangle}$$

$$= \frac{1}{E(z)}$$

So...

$$\frac{1}{E(z)} = z - 1 + \alpha^{-1} - \alpha^{-1} z E(z)$$

self consistent equation for cavity field / Stieltjes transform...

step 2: solving self consistent equation

$$1 = (z - 1 + \bar{\alpha}^1) \varepsilon(z) - \bar{\alpha}^1 z \varepsilon(z)^2$$

$$\bar{\alpha}^1 z \varepsilon^2 - (z + \bar{\alpha}^1 - 1) \varepsilon + 1 = 0$$

ε satisfies a quadratic equation

$$\varepsilon(z) = \frac{(z + \bar{\alpha}^1 - 1) \pm \sqrt{(z + \bar{\alpha}^1 - 1)^2 - 4\bar{\alpha}^1 z}}{2\alpha z}$$

step 3: extract $g(\tau)$ from $\text{Im } \varepsilon(z)$ as $z = \lambda - i\varepsilon \rightarrow 0$

$\varepsilon(\lambda)$ has an imaginary part only when

$$(\lambda + \bar{\alpha}^1 - 1)^2 - 4\bar{\alpha}^1 \lambda < 0$$

quadratic equation

$$\lambda^2 + 2\lambda(\bar{\alpha}^1 - 1) + (\bar{\alpha}^1 - 1)^2 - 4\bar{\alpha}^1 \lambda = 0$$

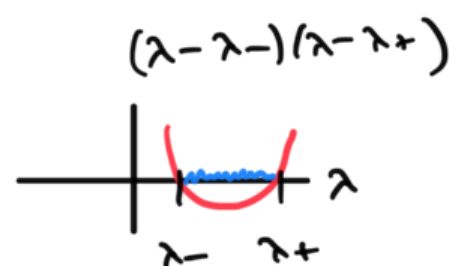
$$\lambda^2 - 2(\bar{\alpha}^1 + 1)\lambda + (\bar{\alpha}^1 - 1)^2 = 0$$

$$\text{roots } \lambda_{\pm} = \frac{2(\bar{\alpha}^1 + 1) \pm \sqrt{4(\bar{\alpha}^1 + 1)^2 - 4(\bar{\alpha}^1 - 1)^2}}{2}$$

$$= (1 + \bar{\alpha}^1) \pm \sqrt{4\bar{\alpha}^1}$$

$$= 1 \pm 2\sqrt{\bar{\alpha}^1} + \bar{\alpha}^1$$

$$\lambda_{\pm} = \left(1 \pm \frac{1}{\sqrt{\alpha}}\right)^2$$



so

$$\varepsilon(z) = \frac{(z - 1 - \alpha) - \sqrt{(z - \lambda_-)(z - \lambda_+)}}{2\alpha z}$$

$$g(\tau) = \lim_{\varepsilon \rightarrow 0} \frac{1}{\pi} \text{Im } \varepsilon(\lambda - i\varepsilon)$$

$$g(\tau) = \begin{cases} \frac{\sqrt{(\lambda - \lambda_-)(\lambda_+ - \lambda)}}{2\alpha\lambda} \\ 0 \end{cases}$$

$$\lambda_- < \lambda < \lambda_+$$

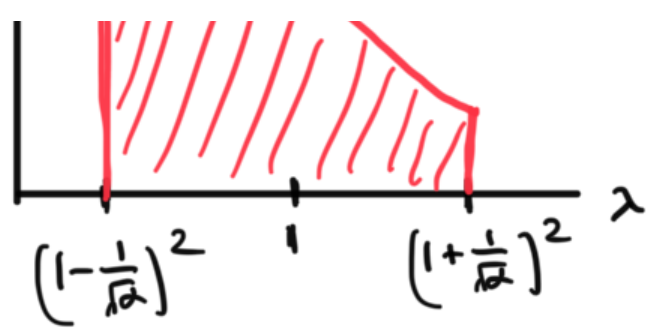
$$\lambda_{\pm} = \left(1 \pm \frac{1}{\sqrt{\alpha}}\right)^2$$

otherwise

$g(\tau)$



• observe more data
of inc



- noise spectrum compresses to 1

Emergent determinism from randomness in high dimensions ??